LAB EXPERIMENT - 11

Department	AIDS
Course Code / Course Name	ITA1409 - ETHICAL HACKING
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Q11. Execute and Display the Results of Nmap Timing and Performance Commands (Aggressive Speed Scan and Insane Speed Scan) in Kali Linux

AIM

To execute and display the results of the Nmap timing and performance commands for aggressive and insane speed scans in the Kali Linux operating system and understand how timing templates affect scan speed and accuracy.

PROCEDURE

1. Open the Kali Linux Terminal and check that Nmap is installed using nmap --version.
2. Select a target that you are authorized to scan (scanme.nmap.org is used here).
3. Execute the Nmap command with the -T4 timing template for the aggressive speed scan.
4. Execute the Nmap command with the -T5 timing template for the insane speed scan.
5. Compare the scan duration and the ports reported by both commands.
6. Observe and verify the output of both commands and understand their functions.
7. Capture screenshots of the terminal showing the execution and output of the commands as evidence.

COMMANDS TO EXECUTE

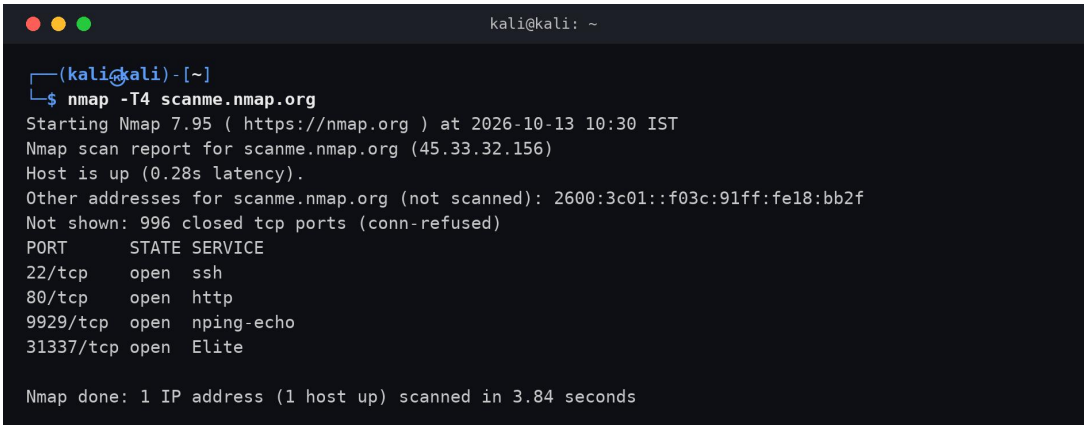
Open the terminal and execute:

1. Aggressive Speed Scan (-T4)

The -T4 (aggressive) timing template speeds up the scan by assuming a fast and reliable network, so it finishes much quicker than the default scan.

```
nmap -T4 scanme.nmap.org
```

OUTPUT:

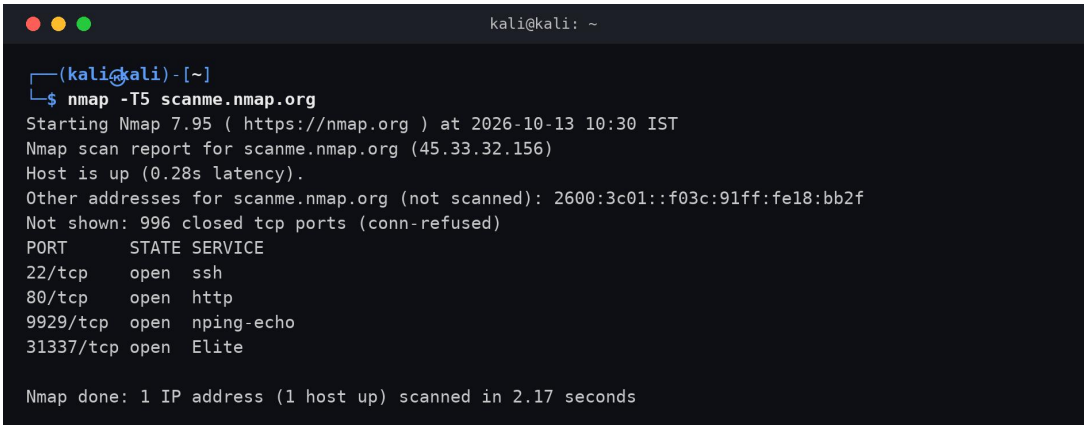


2. Insane Speed Scan (-T5)

The -T5 (insane) timing template is the fastest Nmap speed and sacrifices accuracy for speed, so it may miss open ports on slow networks.

`nmap -T5 scanme.nmap.org`

OUTPUT:



NMAP TIMING TEMPLATES

Template	Name	Description
-T0	Paranoid	Very slow; used for IDS evasion
-T1	Sneaky	Slow; used for IDS evasion
-T2	Polite	Slows down to use less bandwidth and target resources
-T3	Normal	Default speed; balanced scan
-T4	Aggressive	Fast scan on a reliable network
-T5	Insane	Fastest; may miss open ports

OBSERVATION

The aggressive scan (-T4) took about 3.84 seconds, while the insane scan (-T5) finished in about 2.17 seconds because Nmap sends probes with very short timeouts and almost no delay. Faster timing is easily noticed by an IDS and can produce less reliable results.

RESULT

The Nmap timing commands were successfully executed in Kali Linux. The aggressive scan (-T4) completed quickly on a reliable network, while the insane scan (-T5) was the fastest but the least reliable, and both reported the same open ports.